Faculty of Technology & Engineering
Chandubhai S. Patel Institute of Technology

Bachelor of Technology Programme
Computer Science & Engineering (AIML)

# ACADEMIC REGULATIONS & SYLLABUS

(Choice Based Credit System)

Academic Year: 2023-2024

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

**Faculty of Technology and Engineering**

# ACADEMIC REGULATIONS

**Bachelor of Technology (CSE - AIML) Programme**

(Choice Based Credit System)

**Charotar University of Science and Technology (CHARUSAT)**
CHARUSAT Campus, At Post: Changa – 388421, Taluka: Petlad, District: Anand
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**Year – 2023-2024**

# CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

# CHANDUBHAI S PATEL INSTITUTE OF TECHNOLOGY

### COMPUTER SCIENCE & ENGINEERING (AIML)

## Vision

To become the frontiers for dissemination and creation of knowledge in the field of computer science & engineering.

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## Mission

To empower ambitious engineers with expertise and experience to solve the challenges of technological era by:

- Providing diverse interdisciplinary environment to endeavor collaborative research and industry-based projects which offer opportunities for deep-rooted synergy with academia and industry.
- Encouraging experiential learning and self-learning through professional society chapters for better understanding of the subject.
- Enhancing critical and computational thinking to make young minds ready for global platform.

## THE PROGRAMME EDUCATIONAL OBJECTIVES (PEOs)

| | |
|---|---|
| PEO 1 | To establish a strong foundation in students and help them to apply engineering fundamentals, mathematics, science and humanities necessary to formulate, analyze, design and implement engineering solutions. |
| PEO 2 | To engage in **research and find the appropriate solutions for the benefits of mankind** through continuous learning, relearning and unlearning. |
| PEO 3 | To bring out world class engineers, scientists, entrepreneurs, model citizens, problem solvers and **critical thinkers** with skills developed by practices of advance technologies. |
| PEO 4 | To make students aware about their capabilities so that they can analyze real life problems to develop economically viable and socially acceptable solutions that are sustainable using their **technical and analytical** skills. |
| PEO 5 | To provide an environment where the students are nurtured in every aspect for their holistic development such that they achieve excellence in professionalism, **interpersonal skills** as well as moral and ethical conduct. |
| PEO 6 | To infuse in students, the ability to work as a part of team on **multidisciplinary projects** and develop leadership qualities that meet the challenging and changing global needs. |
| PEO 7 | To prepare and make students ready for presenting **themselves on global platform** by encouraging them to adapt rapidly changing industry requirements. |

## PROGRAM OUTCOMES (POs)

At the end of the program, the student will be able to:

| | |
|---|---|
| PO1 | **Engineering knowledge:** Apply knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems. |
| PO2 | **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences. |
| PO3 | **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet specified needs with appropriate consideration for public health and safety, and the cultural, societal, and environmental considerations. |
| PO4 | **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions. |
| PO5 | **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations. |
| PO6 | **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice. |
| PO7 | **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development. |
| PO8 | **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice. |
| PO9 | **Individual and team work**: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings. |
| PO10 | **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions. |
| PO11 | **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one’s own work, as a member and leader in a team, to manage projects and in multidisciplinary environments. |
| PO12 | **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change. |

## PROGRAM SPECIFIC OUTCOMES (PSOs)

At the end of the program, the student will be able to:

| | |
|---|---|
| PSO1 | To instill analytical, technical and ethical skills required to identify optimized IT solutions for complex problems by applying critical and computational thinking. |
| PSO2 | To undergo appropriate research and use innovative technologies for designing IT based solutions that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental benefits. |

# FACULTY OF TECHNOLOGY AND ENGINEERING
## ACADEMIC REGULATIONS
## Bachelor of Technology Programmes
## Choice Based Credit System

To ensure uniform system of education, duration of undergraduate and post graduate programmes, eligibility criteria for and mode of admission, credit load requirement and its distribution between course and system of examination and other related aspects, following academic rules and regulations are recommended.

### *1. System of Education*

Choice based Credit System with Semester pattern of education shall be followed across The Charotar University of Science and Technology (CHARUSAT) both at Undergraduate and Master's levels. Each semester will be at least 90 working day duration. Every enrolled student will be required to take a course works in the chosen subject of specialization and also complete a project/dissertation if any. Apart from the Programme Core courses, provision for choosing University level electives and Programme/Institutional level electives are available under the Choice based credit system.

### *2. Duration of Programme*

| | | |
|---|---|---|
| (i) | Undergraduate programme | (B.Tech) |
| | Minimum | 8 semesters (4 academic years) |
| | Maximum | 16 semesters (8 academic years) |

### *3. Eligibility for admissions*

As enacted by Govt. of Gujarat from time to time.

### *4. Mode of admissions*

As enacted by Govt. of Gujarat from time to time.

### *5. Programme structure and Credits*

***As per annexure – 1 attached***

## 6. *Attendance*

6.1 All activities prescribed under these regulations and listed by the course faculty members in their respective course outlines are compulsory for all students pursuing the courses. No exemption will be given to any student from attendance except on account of serious personal illness or accident or family calamity that may genuinely prevent a student from attending a particular session or a few sessions. However, such unexpected absence from classes and other activities will be required to be condoned by the Dean/Principal.

6.2 Student attendance in a course should be 80%.

## 7 *Course Evaluation*

7.1 The performance of every student in each course will be evaluated as follows:

7.1.1 Internal evaluation by the course faculty member(s) based on continuous assessment, for 30% of the marks for the course; and

7.1.2 Final examination by the University through written paper or practical test or oral test or presentation by the student or a combination of any two or more of these, for 70% of the marks for the course.

*7.2 Internal Evaluation*

***As per Annexure – 1 attached***

*7.3 University Examination*

7.3.1 The final examination by the University for 70% of the evaluation for the course will be through written paper and 100% for practical test or oral test or presentation by the student or a combination of any two or more of these.

**7.3.2 In order to earn the credit in a course a student has to obtain grade other than FF.**

7.4 Performance at Internal & University Examination

7.4.1 Minimum performance with respect to internal marks as well as university examination will be an important consideration for passing a course. Details of minimum percentage of marks to be obtained in the examinations (internal/external) are as follows

| Minimum marks in University Exam per subject | Minimum marks Overall per subject |
|---|---|
| 40% | 45% |

7.4.2 A student failing to score 45% of the final examination will get a FF grade.

7.4.3 If a candidate obtains minimum required marks per subject but fails to obtain minimum required overall marks, he/she has to repeat the university examination till the minimum required overall marks are obtained.

## *8 Grading*

8.1 The total of the internal evaluation marks and final University examination marks in each course will be converted to a letter grade on a ten-point scale as per the following scheme:

***Table: Grading Scheme (UG)***

| **Range of Marks (%)** | **≥80** | **≥73 <80** | **≥66 <73** | **≥60 <66** | **≥55 <60** | **≥50 <55** | **≥45 <50** | **<45** |
|---|---|---|---|---|---|---|---|---|
| **Letter Grade** | **AA** | **AB** | **BB** | **BC** | **CC** | **CD** | **DD** | **FF** |
| **Grade Point** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **0** |

8.2 The student's performance in any semester will be assessed by the Semester Grade Point Average (SGPA). Similarly, his performance at the end of two or more consecutive semesters will be denoted by the Cumulative Grade Point Average (CGPA). The SGPA and CGPA are calculated as follows:

(i) SGPA = $\sum C_i G_i / \sum C_i$ where $C_i$ is the number of credits of course i
$G_i$ is the Grade Point for the course i
and i = 1 to n, n = number of courses in the semester

(ii) CGPA = $\sum C_i G_i / \sum C_i$ where $C_i$ is the number of credits of course i
$G_i$ is the Grade Point for the course i
and i = 1 to n, n = number of courses of all semesters up to which CGPA is computed.

(iii) No student will be allowed to move further if CGPA is less than 3 at the end of every academic year.

(iv) A student will not be allowed to move to third year if he/she has not cleared allthe courses of first year.

(v) A student will not be allowed to move to fourth year if he/she has not clearedall the courses of second year.

## *9. Awards of Degree*

9.1 Every student of the programme who fulfils the following criteria will be eligible for the award of the degree:

9.1.1 He should have earned at least minimum required credits as prescribed in course structure; and

9.1.2 He should have cleared all internal and external evaluation components inevery course; and

9.1.3 He should have secured a minimum CGPA of 4.5 at the end of the programme;

9.14 In addition to above, the student has to complete the required formalities as per the regulatory bodies, if any.

9.2 The student who fails to satisfy minimum requirement of CGPA at the end of program will be allowed to improve the grades so as to secure a minimum CGPA for award of degree. Only latest grade will be considered.

### *10. Award of Class*

The class awarded to a student in the programme is decided by the final CGPA as per the following scheme:

| | |
|---|---|
| Distinction: | CGPA $\geq$ 7.5 & $\leq$10.0 |
| First class: | CGPA$\geq$ 6.0 & < 7.5 |
| Second Class: | CGPA$\geq$ 5.0 & < 6.0 |
| Pass Class: | CGPA < 5.0 |

### *11. Transcript*

The transcript issued to the student at the time of leaving the University will contain a consolidated record of all the courses taken, credits earned, grades obtained, SGPA, CGPA, class obtained, etc.

# CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY (CHARUSAT)

## FACULTY OF TECHNOLOGY & ENGINEERING (FTE)

## CHOICE BASED CREDIT SYSTEM

## FOR

BACHELOR OF TECHNOLOGY & ENGINEERING

**A. Choice Based Credit System:**

With the aim of incorporating the various guidelines initiated by the University Grants Commission (UGC) to bring equality, efficiency and excellence in the Higher Education System, Choice Based Credit System (CBCS) has been adopted. CBCS offers wide range of choices to students in all semesters to choose the courses based on their aptitude and career objectives. It accelerates the teaching-learning process and provides flexibility to students to opt for the courses of their choice and / or undergo additional courses to strengthen their Knowledge, Skills and Attitude.

**1. CBCS – Conceptual Definitions / Key Terms (Terminologies)**

**1.1. Core Courses**

**1.1.1 University Core (UC)**

University Core Courses are those courses which all students of the University of a Particular Level (PG/UG) will study irrespective of their Programme/specialisation.

**1.1.2 Programme Core (PC)**

A 'Core Course' is a course which acts as a fundamental or conceptual base for Chosen Specialisation of Engineering. It is mandatory for all students of a particular Programme and will not have any other choice for the same.

**1.2 Elective Course (EC)**

An 'Elective Course' is a course in which options / choices for course will be offered. It can either be for a Functional Course / Area or Streams of Specialization / Concentration which is / are offered or decided or declared by the University/Institute/Department (as the case may be) from time to time.

**1.2.1 Institute Elective Course (IE)**

Institute Courses are those courses which any students of the University/Institute of a Particular Level (PG/UG) will choose as offered or decided by the University/Institute from time-to-time irrespective of their Programme /Specialisation

**1.2.2 Programme Elective Course (PE):**

A 'Programme Elective Course' is a course for the specific programme in which students will opt for specific course(s) from the given set of functional course/ Area or Streams of Specialization options as offered or decided by the department from time-to-time

**1.2.3 Cluster Elective Course (CE):**

An 'Elective Course' is a course which students can choose from the given set of functional course/ Area or Streams of Specialization options (eg. Common Courses to EC/CE/IT/EE) as offered or decided by the Institute from time-to-time.

## 1.3 Non Credit Course (NC) - AUDIT Course

A ‘Non Credit Course’ is a course where students will receive Participation or Course Completion certificate. This will not be reflected in Student’s Grade Sheet. Attendance and Course Assessment is compulsory for Non Credit Courses

# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

## TEACHING & EXAMINATION SCHEME FOR B TECH PROGRAMME IN CSE ENGINEERING

### CHOICE BASED CREDIT SYSTEM

| Sem | Course Code | Course Title | Teaching Scheme | | | | Examination Scheme | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Contact Hours | | | Credit | Theory | | Practical | | Total |
| | | | Theory | Practical | Total | | Internal | External | Internal | External | |
| Sem 3 | AIML201 | JAVA PROGRAMMING | 2 | 2 | 4 | 3 | 30 | 70 | 25 | 25 | 150 |
| | AMIL202 | DATA COMMUNICATION & NETWORKING | 3 | 2 | 5 | 4 | 30 | 70 | 25 | 25 | 150 |
| | AIML203 | DATA STRUCTURES & ALGORITHMS | 3 | 4 | 7 | 5 | 30 | 70 | 50 | 50 | 200 |
| | AIML204 | INTRODUCTION TO ARTIFICIAL INTELLIGENCE | 3 | 2 | 5 | 4 | 30 | 70 | 25 | 25 | 150 |
| | AIML205 | DATA ANALYSIS USING PYTHON | 2 | 2 | 4 | 3 | 30 | 70 | 25 | 25 | 150 |
| | XXXXX | UNIVERSITY ELECTIVE – I | 2 | | | 2 | 30 | 70 | | | 100 |
| | HS121.02 A | CREATIVITY PROBLEM SOLVING & INNOVATION | 0 | 2 | 2 | 2 | -- | -- | 30 | 70 | 100 |
| | | | **15** | **14** | **27** | **23** | **180** | **420** | **180** | **220** | **1000** |
| Sem 4 | AIML206 | DATABASE MANAGEMENT SYSTEM | 3 | 2 | 5 | 4 | 30 | 70 | 25 | 25 | 150 |
| | AIML207 | DESIGN ANALYSIS OF ALGORITHM | 3 | 2 | 5 | 4 | 30 | 70 | 25 | 25 | 150 |
| | AIML208 | OPERATING SYSTEM | 3 | 2 | 5 | 4 | 30 | 70 | 25 | 25 | 150 |
| | AIML209 | PROBABILISTIC MODELLING AND REASONING WITH PYTHON | 3 | 2 | 5 | 4 | 30 | 70 | 25 | 25 | 150 |
| | AIML210 | R PROGRAMMING FOR DATA SCIENCE AND DATA ANALYSIS | 2 | 2 | 4 | 3 | 30 | 70 | 25 | 25 | 150 |
| | XXXXX | UNIVERSITY ELECTIVE – II | 2 | | | 2 | | | 30 | 70 | 100 |
| | AIML211 | PROJECT-I | 0 | 2 | 2 | 2 | | | 50 | 50 | 100 |

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | HS111.02 A | HUMAN VALUE & PROFESSIONAL ETHICS | 0 | 2 | 2 | 2 | | | 30 | 70 | 100 |
| | | | **16** | **1 4** | **28** | **25** | **150** | **350** | **235** | **315** | **1050** |

| **List of University Electives** | | | |
|---|---|---|---|
| **Code** | **University Elective - I (UE - I)** | **Code** | **University Elective - II (UE - II)** |
| EC281.01 | Introduction to MATLAB Programming | EC282.01 | Prototyping Electronics with Arduino |
| CE281.01 | Art of Programming | CE282.01 | Web Designing |
| CL281.01 | Environmental Sustainability and Climate Change | CL282.01 | Basics of Environmental Impact Assessment |
| EE284 | Python Programming | EE288 | MAINTENANCE OF HOUSEHOLD APPARATUS |
| IT283 | WEB DESIGNING & UI/UX | IT284 | DATA VISULIZATION |
| ME281.01 | Engineering Drawing | ME282.01 | Material Science |
| PH233.01 | Fundamentals of Packaging | PH238.02 | Cosmetics in daily life |
| PD260.01 | Basic Laboratory Techniques | NR261.01 | Life Style Diseases & Management |
| NR251.01 | First Aid & Life Support | PT192.01 | Occupational Health & Ergonomics |
| PT191.01 | Health Promotion and Fitness | CA225 | Programming the Internet |
| CA224 | Introduction to Web Designing | BM241 | Health Care Management |
| BM231 | Banking and Insurance | EE287 | MATLAB PROGRAMMING |
| CL283 | SDG HANDPRINT Laboratory | | |

**Note:**

- **University Elective (UE): -** University Electives are offered in common slots and offered by various departments. Students of any programme can select these electives. Subjects like Research Methodology, Occupational Health & Safety, Engineering Economics, Professional Ethics, and Project Management, Disaster Management, Risk Management etc. can be included.

- **Cluster Elective (CT): -** Institutional Electives means common electives among a cluster of programmes (eg. CE/IT/EC/EE etc.). If Institutional Electives are not applicable, it will be Programme electives
- **Programme Elective (PE): -**
- **Institute Elective (IE):-**
- Provision for Auditing a course will be available
- Audit courses may be offered and decided based on need of the institute/programs.

# B. Tech. Computer Science & Engineering (AIML) Program

# SYLLABI

(Semester – 3)

**CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY**

# AIML201: JAVA PROGRAMMING

**Credits and Hours:**

| Teaching Scheme | Theory | Practical | Tutorial | Total | Credit |
|---|---|---|---|---|---|
| Hours/week | 2 | 4 | - | 7 | **5** |
| Marks | 100 | 100 | - | 200 | |

**Pre-requisite courses:**

- Basic programming skill.

**Outline of the Course:**

| Sr. No. | Title of the unit | Minimum number of hours |
|---|---|---|
| 1. | Fundamental of Programming in Java | 02 |
| 2. | Class Fundamentals | 04 |
| 3. | Array & String Handling | 02 |
| 4. | Inheritance, Interfaces & Packages | 04 |
| 5. | Exceptions Handling | 04 |
| 6. | Multithreaded Programming | 04 |
| 7. | File I/O and NIO | 05 |
| 8. | Java Collection Frameworks and Generics | 05 |
| | Total hours (Theory) : | 30 |
| | Total hours (Lab) : | 60 |
| | Total hours : | 90 |

**Detailed Syllabus:**

| | | | |
|---|---|---|---|
| **1.** | **Fundamental of Programming in Java** | **02 Hours** | **04%** |
| | History of Java, Basic overview of java, Bytecode, JVM Buzz-words, Application and applets, Constants, Variables & Data Types, Comment, Operators, Control Flow | | |
| **2.** | **Class Fundamentals** | **04 Hours** | **09%** |
| | General form of class, Creating class Overloading methods, Constructor, Declaring Object, Returning objects, using objects as parameters, Assigning object reference variables, | | |

| | | | |
|---|---|---|---|
| | Introducing Access control , Understanding static Introducing final, The finalize ( ) method, this keyword ,Garbage collection | | |
| **3.** | **Array & String Handling** | **02 Hours** | **04%** |
| | Array basics, String Array, String class, StringBuffer and StringBuilder class, String Tokenizer Class and Object Class | | |
| **4.** | **Inheritance, Interfaces & Packages** | **04 Hours** | **14%** |
| | Inheritance: Using super creating multilevel Hierarchy, method overriding, Dynamic method dispatch, abstract classes, Using final with Inheritance, Using Package: Defining package, Finding package and CLASSPATH, Access protection, Importing package, Interface: Defining Interface, Default Methods, Implementing Interface, Variables in Interface | | |
| **5.** | **Exceptions Handling** | **04 Hours** | **11%** |
| | Exception types, Try …Catch…Finally, Throw, Throws, creating your own exception subclasses | | |
| **6.** | **Multithreaded Programming** | **04 Hours** | **16%** |
| | Life cycle of thread, thread methods, thread priority, thread exceptions, Implementing Runnable interface, Synchronization and Concurrency | | |
| **7.** | **File NIO** | **05 Hours** | **16%** |
| | File and Directories, Byte streams and character streams, Random Access Files,NIO: Meta Data File Attributes Buffers, Channels, Recursive Operation. | | |
| **8.** | **Collection Framework and Generics** | **05 Hours** | **26%** |
| | Collections of objects, Collections: Sets, Sequence, Map, Understanding Hashing, Use of Array List & Vector, Generics Class, Lamda Expression, Functional Reference, Method Reference, Optional Classes, Processing data with streams | | |

**Course Outcome (COs):**

At the end of the course, the students will be able to

| | |
|---|---|
| CO1 | Understand and implement Object Oriented programming concept using basic syntaxes of control Structures, strings and function for developing skills of logic building activity and garbage collection for saving resources. |
| CO2 | Demonstrate basic problem-solving skills: analyzing problems, modelling a problem as a system of objects, creating algorithms, and implementing models and algorithms in an object-oriented computer language (classes, objects, methods with parameters, abstract classes, interfaces, inheritance and polymorphism). |
| CO3 | Use and develop Concurrency theory: progress guarantees, deadlock, live lock, starvation, linearizability. |
| CO4 | Build and test program using new IO api and exception handling |
| CO5 | Analyze and apply collection framework and generics to solve different data structure algorithms. |
| CO6 | Understand and apply java new features |

**Course Articulation Matrix:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 1 | - | - | - | - | - | - | - | - | - | - | - | - | - |
| CO2 | 2 | 3 | 3 | 2 | - | - | - | - | 1 | - | - | - | 2 | - |
| CO3 | 2 | 3 | 3 | 3 | - | - | - | - | - | - | - | - | 2 | - |
| CO4 | 1 | 2 | 3 | 2 | - | - | - | 1 | 1 | - | - | - | 2 | - |
| CO5 | 1 | 2 | 2 | 2 | - | - | - | - | - | - | - | - | 2 | - |
| CO6 | 1 | 2 | - | - | - | - | - | - | - | - | - | 1 | - | - |

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put "-"

**Recommended Study Material:**

- ❖ **Text Books:**
  1. Java: The Complete Reference, Ninth Edition by Herbert Schildt , Oracle Press.
  2. Java 8 in Action: Lambdas, Streams, and Functional-style Programming by Alan Mycroft and Mario Fusco, Manning Publication

- **Reference Books:**
  1. Thinking in Java, Bruce Eckel, Prentice Hall
  2. Java: A Beginner’s Guide (Sixth Edition) by Herbert Schildt , Oracle Press.
  3. Core Java Volume I--Fundamentals (9th Edition) (Core Series) by Cay S. Horstmann, Prentice Hal

- **Web Materials:**
  1. https://docs.oracle.com

# AIML202: DATA COMMUNICATION & NETWORKING

**Credits and Hours:**

| Teaching Scheme | Theory | Practical | Tutorial | Total | Credit |
|---|---|---|---|---|---|
| Hours/week | 3 | 2 | - | 5 | 4 |
| Marks | 100 | 50 | - | 150 | |

**Pre-requisite courses:**

- N/A

**Outline of the Course:**

| Sr. No. | Title of the unit | Minimum number of hours |
|---|---|---|
| 1. | Introduction | 4 |
| 2. | Physical Layer | 4 |
| 3. | Data Link Layer | 10 |
| 4. | Medium Access Sub-layer | 6 |
| 5. | Network Layer | 12 |
| 6. | Transport Layer | 6 |
| 7. | Application Layer | 3 |
| | Total hours (Theory) : | 45 |
| | Total hours (Lab) : | 30 |
| | Total hours : | 75 |

**Detailed Syllabus:**

| 1. | **Introduction and Basic Concepts** | **04 Hours** | **10%** |
|---|---|---|---|
| | Introduction to Computer Networks, Network Architecture, and OSI reference model, services, network standardization. | | |
| **2.** | **Physical Layer** | **04 Hours** | **10%** |
| | Basics of data communication, Guided Transmission Media, Wireless Transmission, Switching | | |

| | | | |
|---|---|---|---|
| **3.** | **Data Link Layer** | **10 Hours** | **20%** |
| | Design issues, Error detection and correction, Elementary Data Link Protocols, Sliding window protocols, Data Link Protocols in Internet | | |
| **4.** | **Medium Access Sublayer** | **06 Hours** | **10%** |
| | Channel Allocation, Multiple Access Protocols, IEEE 802 standards for Ethernet LAN, Wireless LANS, Broadband Wireless, Data Link Layer Switching | | |
| **5.** | **Network Layer** | **12 Hours** | **20%** |
| | Design issues, Routing algorithms, Congestion control, Internetworking-Fragmentation, Addressing, IPv4, IPv6, ARP, DHCP, ICMP, Network layer in the Internet | | |
| **6.** | **Transport Layer** | **06 Hours** | **10%** |
| | Design issues, Transport Service, Elements of Transport Protocol, Internet Transport Protocols – UDP & TCP | | |
| **7.** | **Application Layer** | **03 Hours** | **10%** |
| | Domain Name System, Electronic mail, WWW | | |

**Course Outcome (COs):**

At the end of the course, the students will be able to

| | |
|---|---|
| CO1 | Understand and identify different physical layer transmission fundamentals such as types of signals, transmission, multiplexing, types of medium and modulation. |
| CO2 | Evaluate existing layer-2 networking standards and implementations. |
| CO3 | Evaluate key networking protocols, and their hierarchical relationship in the context of a conceptual model, such as the OSI and TCP/IP framework. |
| CO4 | Understand existing different medium access protocols and evaluate for adoption for future networking. |
| CO5 | Understand and differentiate functionality of existing network routing protocols. |
| CO6 | Measure different network parameter such as Throughput & different types of delays. |

**Course Articulation Matrix:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 2 | 1 | - | 1 | - | - | - | - | - | - | - | 2 | - | - |

| | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO 2 | 1 | 2 | 1 | 1 | 2 | - | - | - | - | - | - | 2 | 1 | 1 |
| CO 3 | 3 | 2 | 2 | 1 | 2 | - | - | - | - | 1 | - | 2 | 1 | 1 |
| CO 4 | 2 | 3 | 1 | 1 | 1 | - | - | - | - | 2 | - | 2 | - | - |
| CO 5 | 3 | 2 | 3 | 1 | 3 | - | - | - | 1 | 1 | - | 3 | 1 | 1 |
| CO 6 | 3 | 3 | 3 | 2 | 3 | - | - | - | 2 | 2 | - | 3 | 3 | 2 |

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

**Recommended Study Material:**

❖ **Text book:**

1. Data communication & Networking, Behrouz Forouzan, McGraw-Hill

❖ **Reference book:**

1. Data and Computer Communications, William Stallings, Prentice Hall
2. Computer Network, Andrew S. Tanenbaum, Fourth Edition, Prentice Hall

❖ **Web material:**

1. www.wikipedia.org
2. http://www.webopedia.com

❖ **Software:**

1. Wireshark
2. Cisco Packet Tracer

# AIML203: DATA STRUCTURE & ALGORITHMS

**Credits and Hours:**

| Teaching Scheme | Theory | Practical | Tutorial | Total | Credit |
|---|---|---|---|---|---|
| Hours/week | 3 | 4 | - | 7 | **5** |
| Marks | 100 | 100 | - | 200 | |

**Pre-requisite courses:**

- Programming Language

**Outline of the Course:**

| Sr. No. | Title of the unit | Minimum number of hours |
|---|---|---|
| 1. | Introduction to Data Structure | 04 |
| 2. | Linear Data Structure | 12 |
| 3. | Non-Linear Data Structure | 16 |
| 4. | Sorting | 10 |
| 5. | Searching | 03 |
| | Total hours (Theory) : | 45 |
| | Total hours (Lab) : | 30 |
| | Total hours : | 75 |

**Detailed Syllabus:**

| 1. | **Introduction** | **04 Hours** | **08%** |
|---|---|---|---|
| | Introduction to data structure (Types of data structure), Introduction to algorithms. Algorithm Analysis and Big O notation, Memory representation of Array: Row Order and Column Order, Abstract Data Types (ADT) | | |
| **2.** | **Linear Data Structure** | **12 Hours** | **27%** |
| | **Stack**: Operations: push, pop, peep, change, Applications of Stack: Recursion: Recursive Function Tracing, Principles of recursion, Tail recursion, Removal of Recursion, Tower of Hanoi, Conversion: Infix to Postfix, Infix to Prefix. Evaluation: Prefix and Postfix expression, | | |

| | | | |
|---|---|---|---|
| | **Queue** Simple Queue: Insert and Delete operation, Circular Queue: Insert and Delete operation, Concepts of: Priority Queue, Double-ended Queue, Applications of Queue,<br>**Linked List:** Memory Representation of LL, Singly Linked List, Doubly Linked List, Circular Linked List, Applications of Linked List | | |
| **3.** | **Non-Linear Data Structure** | **16 Hours** | **36%** |
| | **Tree**: Tree Concepts, Tree Traversal Techniques: Pre-order, Post-order and In-order (Recursive and Iterative), Binary Search Tree: Iterative and Recursive, Balanced Trees (AVL Trees, Applications of Tree, Skip list<br>**Heaps**: Priority queues and Binary Heaps<br>**Graph:** Graph concepts, Memory Representation of Graph, BFS and DFS, Applications of Graph | | |
| **4.** | **Sorting** | **10 Hours** | **23%** |
| | Sorting (concepts, Selection Sort, Bubble Sort, Merge Sort, Radix Sort, Insertion Sort, Heap Sort, Quick Sort, Counting sort, Topological sort) | | |
| **5.** | **Searching** | **03 Hours** | **06%** |
| | Sequential Search, Binary Search, Hashing, Hashing Functions, Collision-Resolution Techniques, Applications | | |

**Course Outcome (COs):**

At the end of the course, the students will be able to:

| | |
|---|---|
| CO1 | Understand and Implement Algorithms and core Data Structures such as stack, queue, hash table, priority queue, binary search tree and graph in programming language. |
| CO2 | Analyse data structures in storage, retrieval and computation of ordered or unordered data. |
| CO3 | Compare alternative implementations of data structures with respect to demand and performance. |
| CO4 | Describe and evaluate the properties, operations, applications, strengths and weaknesses of different data structures. |

| | |
|---|---|
| CO5 | Apply and select the most suitable data structures to solve programming challenges. |
| CO6 | Discover advantages and disadvantages of specific algorithms. |

**Course Articulation Matrix:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 3 | - | 2 | - | - | - | - | - | - | - | - | - | 3 | - |
| CO2 | - | 2 | - | - | - | - | - | - | - | - | - | - | 3 | - |
| CO3 | - | 3 | 3 | 3 | - | - | - | - | - | - | - | - | 2 | - |
| CO4 | - | 1 | - | 1 | 2 | - | - | - | - | - | - | - | 2 | - |
| CO5 | 2 | 2 | 2 | 2 | - | - | - | - | - | - | - | 2 | 3 | - |
| CO6 | 2 | - | - | - | - | - | - | - | - | - | - | - | 2 | - |

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

**Recommended Study Material:**

❖ **Text book:**

1. An Introduction to Data Structures with Applications, Jean-Paul Tremblay, Paul G. Sorenson, McGraw-Hill.
2. Data structure with C, Lipschutz, TMH
3. Introduction to Algorithms: Cormen, Leiserson, Rivest and Stein: Prentice Hall of India
4. Data Structures and Algorithms: Aho, Hopcroft and Ullmann: Addison Wesley.

❖ **Reference book:**

1. Classic Data structures, D.Samanta, Prentice-Hall International.0
2. Data Structures using C & C++, Ten Baum, Prentice-Hall International.
3. Data Structures: A Pseudo-code approach with C, Gilberg & Forouzan, Thomson Learning.

4. Fundamentals of Data Structures in C++, Ellis Horowitz, Sartaj Sahni, Dinesh Mehta, W. H. Freeman.
5. "A Practical Introduction to Data Structures and Algorithm Analysis" by Clifford A. Shaffer
6. Data Structures and Algorithm in Java: Goodrich and Tamassia: John Wiley and Sons.

- **Web material:**
    1. http://www.leda-tutorial.org/en/official/ch02s02s03.html
    2. http://www.leda-tutorial.org/en/official/ch02s02s03.html
    3. http://www.softpanorama.org/Algorithms/sorting.shtml

# AIML204: Introduction to AI and Data Science

**Credits and Hours:**

| Teaching Scheme | Theory | Practical | Tutorial | Total | Credit |
|---|---|---|---|---|---|
| Hours/week | 3 | 2 | - | 5 | **4** |
| Marks | 100 | 50 | - | 150 | |

**Pre-requisite courses:**

- Programming Language

**Outline of the Course:**

| Sr. No. | Title of the unit | Minimum number of hours |
|---|---|---|
| 1. | Introduction to Data Science | 05 |
| 2. | Data Science Process | 12 |
| 3. | Introduction to Machine Learning | 10 |
| 4. | Introduction to AI | 10 |
| 5. | Introduction to Data Analytics | 08 |
| | Total hours (Theory) : | 45 |
| | Total hours (Lab) : | 30 |
| | Total hours : | 75 |

**Detailed Syllabus:**

| **1.** | **Introduction to Data Science** | **05 Hours** | **12%** |
|---|---|---|---|
| | Defining Data Science and Big Data, Benefits and Uses of Data Science and Big Data, Facets of Data, Structured Data, Unstructured Data, Natural Language, Machine generated Data, Graph based or Network Data, Audio, Image, Video, Streaming data, Data Science Process, Big data ecosystem and data science, distributed file systems, Distributed programming framework, data integration framework, machine learning framework, No SQL Databases, scheduling tools, benchmarking tools, system deployments | | |
| **2.** | **Data Science Process** | **12 Hours** | **27%** |

| | | | |
|---|---|---|---|
| | Six steps of data science processes, define research goals, data retrieval, cleansing data, correct errors as early as possible, integrating – combine data from different sources, transforming data, exploratory data analysis, Data modelling, model and variable selection, model execution, model diagnostic and model comparison, presentation and automation. | | |
| **3.** | **Introduction to Machine Learning** | **10 Hours** | **23%** |
| | What is Machine Learning, Learning from Data, History of Machine Learning, Big Data for Machine Learning, Leveraging Machine Learning, Descriptive vs Predictive Analytics, Machine Learning and Statistics, Artificial Intelligence and Machine Learning, Types of Machine Learning – Supervised, Unsupervised, Semi-supervised, Reinforcement Learning, Types of Machine Learning Algorithms, Classification vs Regression Problem, Bayesian, Clustering, Decision Tree, Dimensionality Reduction, Neural Network and Deep Learning, Training machine learning systems | | |
| **4.** | **Introduction to AI** | **10 Hours** | **23%** |
| | What is AI, Turing test, cognitive modelling approach, law of thoughts, the relational agent approach, the underlying assumptions about intelligence, techniques required to solve AI problems, level of details required to model human intelligence, successfully building an intelligent problem, history of AI | | |
| **5.** | **Introduction to Data Analytics** | **08 Hours** | **15%** |
| | Working with Formula and Functions, Introduction to Power BI & Charts, Logical functions using Excel, Analysing Data with Excel. | | |

**Course Outcome (COs):**

At the end of the course, the students will be able to:

| | |
|---|---|
| CO1 | Understand the basic concepts of machine learning and data science with typical applications |
| CO2 | Understanding how to build and validate models and improve them iteratively |
| CO3 | Understand the core concepts of artificial intelligence and applications |
| CO4 | Apply knowledge representation with artificial intelligence using FOL and Predicate logic. |

**Course Articulation Matrix:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 3 | - | 2 | - | - | - | - | - | - | - | - | - | 3 | - |
| CO2 | - | 2 | - | - | - | - | - | - | - | - | - | - | 3 | - |
| CO3 | - | 3 | 3 | 3 | - | - | - | - | - | - | - | - | 2 | - |
| CO4 | - | 1 | - | 1 | 2 | - | - | - | - | - | - | - | 2 | - |
| CO5 | 2 | 2 | 2 | 2 | - | - | - | - | - | - | - | 2 | 3 | - |
| CO6 | 2 | - | - | - | - | - | - | - | - | - | - | - | 2 | - |

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put "-"

**Recommended Study Material:**

- **Text book:**
  1. Artificial Intelligence 3e: A Modern Approach Paperback – By Stuart J Russell & Peter Norvig; Publisher – Pearson
- **Reference book:**
  1. Artificial Intelligence Third Edition By Kevin Knight, Elaine Rich, B. Nair – McGrawHill\
  2. Artificial Intelligence Third Edition By Patrick Henry Winston – Addison-Wesley Publishing Company

- **Web material:**
  1. https://www.heavy.ai/learn/data-science
  2. https://www.geeksforgeeks.org/data-science-process/
  3. https://monkeylearn.com/machine-learning/
  4. https://www.harness.io/blog/ai-ml-introduction

# AIML205: Data Analysis using Python

**Credits and Hours:**

| Teaching Scheme | Theory | Practical | Tutorial | Total | Credit |
|---|---|---|---|---|---|
| Hours/week | 2 | 2 | - | 4 | **3** |
| Marks | 100 | 50 | - | 150 | |

**Pre-requisite courses:**

- Programming Language

**Outline of the Course:**

| Sr. No. | Title of the unit | Minimum number of hours |
|---|---|---|
| 1. | Python Programming Basic | 04 |
| 2. | Data Structure, functions, files | 11 |
| 3. | Numpy: Array and Vectorized Computation | 10 |
| 4. | Pandas | 02 |
| 5. | Visualization with Matplotlib And Plotting with Pandas and Seaborn | 03 |
| | Total hours (Theory) : | 30 |
| | Total hours (Lab) : | 30 |
| | Total hours : | 60 |

**Detailed Syllabus:**

| | | | |
|---|---|---|---|
| **1.** | **Python Programming Basics** | **04 Hours** | **13%** |
| | Python interpreter, IPython Basics, Tab completion, Introspection, %run command, magic commands, matplotlib integration, python programming, language semantics, scalar types. Control flow. | | |
| **2.** | **Data Structure, functions, files** | **11 Hours** | **36%** |
| | tuple, list, built-in sequence function, dict, set, functions, namescape, scope, local function, returning multiple values, functions are objects, lambda functions, error and exception handling, file and operation systems | | |

| | | | |
|---|---|---|---|
| **3.** | **Numpy: Array and Vectorized Computation** | **10 Hours** | **33%** |
| | Multidimensional array object. Creating arrays, arithmetic with numpy array, basic indexing and slicing, Boolean indexing, transposing array and swapping axes, universal functions, array-oriented programming with arrays, conditional logic as arrays operations, file input and output with array | | |
| **4.** | **Pandas** | **02 Hours** | **8%** |
| | Pandas data structure, series, DataFrame, Index Object, Reindexing, dropping entities from an axis, indexing, selection and filtering, integer indexes, arithmetic and data alignment, function application and mapping, scoring and ranking, correlation and covariance, unique values, values controls and membership, reading and writing data in text format | | |
| **5.** | **Visualization with Matplotlib And Plotting with Pandas and Seaborn** | **03 Hours** | **10%** |
| | Figures and subplots, colors, markers, line style, ticks, labels, legends, annotation and drawing on sublots, matplotlib configuration.<br>line plots, bar plots, histogram, density plots, scatter and point plots, facet grids and categorical data | | |

**Course Outcome (COs):**

At the end of the course, the students will be able to:

| | |
|---|---|
| CO1 | Understand the basics of algorithm building for computing and programming. |
| CO2 | Understand the basics of python programming language. |
| CO3 | Apply modularization techniques for problem solving through python. |
| CO4 | Outline the concepts of Lists, Dictionaries and Files. |

**Course Articulation Matrix:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 3 | - | 2 | - | - | - | - | - | - | - | - | - | 3 | - |
| CO2 | - | 2 | - | - | - | - | - | - | - | - | - | - | 3 | - |
| CO3 | - | 3 | 3 | 3 | - | - | - | - | - | - | - | - | 2 | - |
| CO4 | - | 1 | - | 1 | 2 | - | - | - | - | - | - | - | 2 | - |
| CO5 | 2 | 2 | 2 | 2 | - | - | - | - | - | - | - | 2 | 3 | - |
| CO6 | 2 | - | - | - | - | - | - | - | - | - | - | - | 2 | - |

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put "-"

**Recommended Study Material:**

- ❖ **Text book:**
  1. Achim Klenke, (2014), Probability Theory A Comprehensive Course Second Edition, Springer, ISBN 978-1-4471-5360-3

- ❖ **Reference book:**
  1. Christian Heumann, Michael Schomaker Shalabh (2016), Introduction to Statistics and Data Analysis with Exercises, Solutions and Applications in R, Springer International Publishing, ISBN 978-3-319-46160-1
  2. Douglas C. Montgomery, (2012), Applied Statistics and Probability for Engineers, 5th Edition,, Wiley India, ISBN: 978-8-126-53719-8.

- ❖ **Web material:**

1. https://www.guru99.com/python-tutorials.html
2. https://www.oreilly.com/library/view/python-for-data/9781449323592/ch04.html
3. https://machinelearningmastery.com/data-visualization-in-python-with-matplotlib-seaborn-and-bokeh/
4. https://jovian.com/learn/data-analysis-with-python-zero-to-pandas/lesson/lesson-5-data-visualization-with-matplotlib-and-seaborn

# B. Tech. Computer Science & Engineering (AIML) Program

# SYLLABI

(Semester – 4)

**CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY**

# AIML206: DATABASE MANAGEMENT SYSTEM

**Credits and Hours:**

| Teaching Scheme | Theory | Practical | Tutorial | Total | Credit |
|---|---|---|---|---|---|
| Hours/week | 3 | 4 | - | 7 | **5** |
| Marks | 100 | 50 | - | 150 | |

**Pre-requisite courses:**

- Data Structure

**Outline of the Course:**

| Sr. No. | Title of the unit | Minimum number of hours |
|---|---|---|
| 1. | Introductory concepts of DBMS | 03 |
| 2. | Relational Model | 05 |
| 3. | Entity-Relationship model | 07 |
| 4. | Formal Relational Query Languages | 05 |
| 5. | Relational Database Design | 07 |
| 6. | Transaction & Recovery Management | 08 |
| 7. | Advanced Transaction Processing | 06 |
| 8. | Database Security | 06 |
| 9. | Indexing and Hashing | 05 |
| 10. | Query Processing & Query Optimization | 05 |
| | Total hours (Theory) : | 45 |
| | Total hours (Lab) : | 60 |
| | Total hours : | 105 |

**Detailed Syllabus:**

| | | | |
|---|---|---|---|
| **1.** | **Introductory concepts of DBMS** | **03 Hours** | **06%** |
| | Introduction and applications of DBMS, Purpose of database, Data Independence, Database System architecture- levels, Mappings, Database users and DBA | | |
| **2.** | **Relational Model** | **05 Hours** | **15%** |

| | | | |
|---|---|---|---|
| | Structure of Relational Databases, Database Schema, Schema Diagram, Domains, Relations, Relational Query Languages, Relational Operations | | |
| **3.** | **Entity-Relationship model** | **07 Hours** | **15%** |
| | Basic concepts, Design process, Constraints, Keys, Design issues, E-R diagrams, Weak Entity Sets, Extended E-R features- Generalization, Specialization, Aggregation, Reduction to E-R database schema | | |
| **4.** | **Formal Relational Query Languages** | **05 Hours** | **11%** |
| | The relational Algebra, The Tuple Relational Calculus, The Domain Relational Calculus | | |
| **5.** | **Relational Database design** | **07 Hours** | **13%** |
| | Functional Dependency–definition, Trivial and Non-Trivial FD, Closure of FD set, Closure of attributes, Irreducible set of FD, Normalization – 1NF, 2NF,3NF, Decomposition using FD- Dependency Preservation | | |
| **6.** | **Transaction & Recovery Management** | **08 Hours** | **15%** |
| | Transaction concepts, Properties of Transactions, Serializability of transactions, Testing for Serializability, System recovery, Two- Phase Commit protocol, Recovery and Atomicity, Log-based recovery¸ Concurrent executions of transactions and related problems, Locking mechanism, Solution to Concurrency Related Problems, Deadlock, Two-phase locking protocol, Intent locking | | |
| 7. | **Advanced Transaction Processing** | **06 Hours** | **10%** |
| | Transaction-Processing Monitors, Transactional Workflows, Main-Memory Databases, Real-Time Transaction Systems, Long-Duration Transactions | | |
| **8.** | **Database Security** | **06 Hours** | **10%** |
| | Views - What are views for? View retrievals, View updates, Snapshots (a digression), Materialized view, Security – Security and Authentication, authorization in SQL, Data | | |

| | | | |
|---|---|---|---|
| | encryption, Missing Information - An overview of the 3VL approach | | |
| **9.** | **Indexing and Hashing** | **05 Hours** | **15%** |
| | Basic Concepts, Ordered Indices, B+-Tree Index Files, B+-Tree Extensions, Multiple-Key Access, Static Hashing, Dynamic Hashing, Comparison of Ordered Indexing and Hashing, Bitmap Indices, Index Definition in SQL | | |
| **10.** | **Query Processing & Query Optimization** | **05 Hours** | **10%** |
| | Overview, Measures of Query Cost, Selection Operation, Sorting, Join, Evaluation of Expressions, Transformation of relational Expressions, Estimating Statistics of expression results, Query Evaluation plans | | |

**Course Outcomes (COs):**

At the end of the course, students will be able to

| | |
|---|---|
| CO1 | Apply the concepts of engineering i.e collecting data, organize the data in the systematic form, and arrange the data in a computational way and applying mathematics formation. |
| CO2 | Analyse how data are stored and maintained using data models. Ready to assimilate the concept of data abstraction and design queries using SQL. Identify how data is represented in the relational model and create relations using SQL language |
| CO3 | Identify and evaluate the constructs in the E-R model and issues involved in developing an E-R diagram. Convert an E-R diagram into a relational database schema. Declare and enforce integrity constraints on database using a state-of-art RDBMS. |
| CO4 | Produce aggregate operators to write SQL queries which are not expressible in relational algebra. “More mathematical” notation may apply and also used in research and other venues. Combining these concepts allows production of sophisticated queries. |
| CO5 | Decompose un-normalized tables into normalized compliant tables. Design and implement a normalize database schema for a given problem-domain. Produce strategies to minimise risks of security breaches in a range of network |

| | |
|---|---|
| | environments and data storage systems. Compute retrieval time and concluding with suitable indexing technique. |
| CO6 | Compare transactions and their properties with (ACID) and without ACID. Apply locking protocol to ensure isolation. Develop logging technique to ensure atomicity and durability. Design a logical view which can be used for analytical tasks. Develop practical experience of the design and implement scalable, secure databases. |

**Course Articulation Matrix:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 3 | 3 | 2 | 2 | 1 | 2 | 1 | 1 | - | - | - | 1 | 2 | - |
| CO2 | 3 | 3 | 3 | 2 | 3 | 1 | 2 | - | - | - | - | 3 | 2 | 1 |
| CO3 | 3 | 3 | 3 | 3 | 2 | 2 | 1 | 3 | 2 | 1 | - | 2 | 3 | 2 |
| CO4 | 3 | 3 | 1 | 3 | 1 | 1 | - | 2 | 2 | - | - | 2 | 2 | 2 |
| CO5 | 3 | 3 | 3 | 2 | 3 | 1 | - | 2 | 2 | 1 | - | 3 | 2 | 2 |
| CO6 | 3 | 3 | 2 | 2 | - | 2 | 1 | 3 | 1 | - | - | 3 | 3 | 1 |

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

**Recommended Study Material:**

- ❖ **Text book:**
  1. Database System Concepts, Abraham Silberschatz, Henry F. Korth& S. Sudarshan, McGraw Hill.
  2. An introduction to Database Systems, C J Date, Addition-Wesley
- ❖ **Reference book:**
  1. “Fundamentals of Database Systems”, R. Elmasri and S.B. Navathe, the Benjamin / Cumming Pub. Co
  2. SQL,PL/SQL the Programming Language of oracle, Ivan Bayross, BPB Publications

3. Oracle: The Complete Reference, George Koch, Kevin Loney, TMH /oracle press

❖ **Web material:**

1. http://www.sql.org
2. http://www.w3schools.com
3. http://www.sqlcourse.com

❖ **Software:**

1. Oracle 10g
2. SQL Lite
3. Live SQL
4. Firebase
5. Squirrel SQL
6. Postgre SQL

# AIML207: DESIGN & ANALYSIS OF ALGORITHMS

**Credits and Hours:**

| Teaching Scheme | Theory | Practical | Tutorial | Total | Credit |
|---|---|---|---|---|---|
| Hours/week | 4 | 2 | - | 6 | **5** |
| Marks | 100 | 50 | - | 150 | |

**Pre-requisite courses:**

- Data Structure and Algorithms
- Programming language
- Discrete Mathematics

**Outline of the Course:**

| Sr. No. | Title of the unit | Minimum number of hours |
|---|---|---|
| 1. | Introduction | 14 |
| 2. | Greedy Algorithm | 16 |
| 3. | Dynamic Programming | 14 |
| 4. | Divide and Conquer Algorithm | 06 |
| 5. | Searching Algorithms and String Matching Algorithms | 06 |
| 6. | Complexity Theory | 04 |
| | Total hours (Theory) : | 60 |
| | Total hours (Lab) : | 30 |
| | Total hours : | 90 |

**Detailed Syllabus:**

| **1.** | **Introduction** | **14 Hours** | **24%** |
|---|---|---|---|
| | Introduction to Algorithms, Analysis and Design Techniques of Algorithm, Model for Analysis of Algorithm: Random Access Machine (RAM), Algorithm Analysis Techniques: Mathematics, Empirical and Asymptotic Analysis: Big-O, Big-Theta and Big-Omega, worst, average and best case analysis Complexity Analysis of Iterative Algorithm: Primitive Operations, Analysing control statements | | |

| | | | |
|---|---|---|---|
| | Complexity Analysis techniques for Recurrence Relations: substitution method, Iterative Method, Recursion Tree Method, Master's Theorem<br>Basic Sorting Algorithms with best case, average case and worst case analysis: Selection Sort, Insertion Sort, Bubble Sort | | |
| **2.** | **Greedy Design Technique** | **16 Hours** | 26% |
| | Greedy choice and optimal substructure property of Greedy approach, Problem Solving using greedy: making change, fractional knapsack, Job Scheduling, Huffman coding, Graph Coloring Algorithm. Minimum spanning trees -Prims and Kruskal algorithm. Graph algorithms: Breath First Search, Depth First Search, Shortest Path Algorithms: Dijkstra's algorithm, Floyd-Warshall Algorithm, Bellman-Ford Algorithm. | | |
| 3. | **Dynamic Programming** | **14 Hours** | 24% |
| | Dynamic Programming: The Principle of Optimality, Dynamic Programming using memorization and bottom up approach, problem solving using Dynamic Programming: Calculating the Binomial Coefficient, Assembly Line-Scheduling, Knapsack Problem, Matrix Chain Multiplication, Subset Sum Problem, Longest Common Subsequence | | |
| **4.** | **Divide and Conquer Approach** | **06 Hours** | **10%** |
| | Binary Search, Sorting Algorithms with best, average and worst case analysis: Merge Sort and Quick Sort, Matrix Multiplication | | |
| **5.** | **Searching Algorithms and String Matching Algorithms** | **06 Hours** | **10%** |
| | **Backtracking** –The Knapsack Problem; The Eight Queens problem, Sum of Subset Problem<br>**Branch and Bound** –The Assignment Problem, The Knapsack Problem<br>**String Matching Algorithms:** The naïve string-matching algorithm, The Rabin-Karp algorithm, KMP Algorithm for Pattern Searching, Boyer–Moore string-search algorithm | | |

| 6. | **Complexity Theory** | **4 Hours** | **6%** |
|---|---|---|---|
| | The class P and NP Problem, Polynomial reduction, NP-Completeness Problem, NP-Hard problems | | |

**Course Outcome (COs):**

At the end of the course, the students will be able to

| | |
|---|---|
| CO1 | Analyse how to analyse algorithms and estimate their best case, worst-case and average-case behaviour (in easy cases) |
| CO2 | Describe the greedy paradigm and explain when an algorithmic design situation calls for it. Recite algorithms that employ this paradigm and analyse them. |
| CO3 | Describe the dynamic-programming paradigm and explain when an algorithmic design situation calls for it. Recite algorithms that employ this paradigm and analyse them. |
| CO4 | Describe the divide-and-conquer paradigm and explain when an algorithmic design situation calls for it. Recite algorithms that employ this paradigm. Derive and solve recurrences describing the performance of divide-and-conquer algorithms. |
| CO5 | Understand and apply various graph algorithms for finding shorted path and minimum spanning tree. |
| CO6 | Understand and apply string matching algorithms for finding the pattern from the text. |
| CO6 | Understand a problem as computationally tractable or intractable, and discuss strategies to address intractability. |

**Course Articulation Matrix:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 2 | 1 | - | - | - | - | - | - | - | - | - | - | 1 | - |
| CO2 | 2 | 2 | - | - | - | - | - | - | - | - | - | 2 | 2 | - |
| CO3 | 3 | 3 | 3 | 3 | 2 | - | - | - | - | - | - | 2 | 2 | - |
| CO4 | 2 | 3 | 3 | 1 | - | - | - | - | - | - | - | - | 2 | - |

| | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO5 | 1 | - | 1 | - | - | - | - | - | - | - | - | 2 | 1 | 1 |
| CO6 | 3 | 1 | - | - | - | - | - | - | - | - | - | - | 1 | - |

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put "-"

**Recommended Study Material:**

❖ **Text book:**

1. Introduction to Algorithms by Thomas H. Cormen, Charles E. Leiserson, Ronald Rivest and Clifford Stein, MIT Press

❖ **Reference book:**

1. Fundamental of Algorithms by Gills Brassard, Paul Bratley, Pentice Hall of India.
2. Fundamental of Computer Algorithms by Ellis Horowitz, Sartazsahni and sanguthevar Rajasekarm, Computer Sci.P.
3. Design & Analysis of Algorithms by P H Dave & H B Dave, Pearson Education.

❖ **Web material:**

1. http://highered.mcgraw-hill.com/sites/0073523402/

**CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY**
**FACULTY OF TECHNOLOGY&ENGINEERING**
**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING**

# AIML350: OPERATING SYSTEM

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## Credits and Hours:

| Teaching Scheme | Theory | Practical | Tutorial | Total | Credit |
|---|---|---|---|---|---|
| Hours/week | 3 | 2 | - | 5 | 4 |
| Marks | 100 | 50 | - | 150 | |

**Pre-requisite courses:**

- Introduction to computer and computer architecture.

## Outline of the course:

| Sr. No. | Title of the unit | Minimum number of hours |
|---|---|---|
| 1. | Introduction | 02 |
| 2. | Process Management | 04 |
| 3. | Inter process Communication | 08 |
| 4. | Deadlock | 06 |
| 5. | Memory Management | 08 |
| 6. | Input Output Management | 07 |
| 7. | File Systems | 06 |
| 8. | Unix/Linux File System | 04 |

**Total hours (Theory): 45**

**Total hours (Lab): 30**

**Total hours: 75**

**Detailed Syllabus:**

| | | | |
|---|---|---|---|
| **1.** | **Introduction** | **02 Hours** | **05%** |
| 1.1 | What is an OS? Evolution of OS | | |
| 1.2 | OS Services | | |
| 1.3 | Types of OS | | |
| 1.4 | Concepts of OS | | |
| 1.5 | Different Views Of OS | | |
| **2.** | **Process Management** | **04 Hours** | **08%** |
| 2.1 | Process, Process Control Block, Process States, | | |
| 2.2 | Threads, Types of Threads and Dispatching, Concurrent Threads | | |
| **3.** | **Inter process Communication** | **08 Hours** | **15%** |
| 3.1 | Race Conditions, Critical Section, Co-operating Thread/Mutual Exclusion | | |
| 3.2 | Hardware Solution, Strict Alternation, Peterson's Solution | | |
| 3.3 | The Producer Consumer Problem, Semaphores, Event Counters, | | |
| | Monitors | | |
| 3.4 | Message Passing and Classical IPC Problems: Reader's & Writer Problem, Dinning Philosopher Problem. | | |
| **4.** | **Deadlock** | **06 Hours** | **15%** |
| 4.1 | Deadlock Problem, Deadlock Characterization | | |
| 4.2 | Deadlock Detection, Deadlock recovery | | |
| 4.3 | Deadlock avoidance: Banker's algorithm for single & multiple resources | | |
| 4.4 | Deadlock Prevention. | | |
| 4.5 | CPU Scheduling, Protection: Address space, Address Translation | | |
| **5.** | **Memory Management** | **08 Hours** | **18%** |
| 5.1 | Paging: Principle of Operation, Page Allocation, H/W Support For Paging | | |
| 5.2 | Multiprogramming with Fixed partitions | | |
| 5.3 | Segmentation | | |
| 5.4 | Swapping | | |
| 5.5 | Virtual Memory: Concept, Performance Of Demand Paging, Page | | |
| | Replacement Algorithms, Thrashing and Working Sets | | |
| **6.** | **Input Output Management** | **07 Hours** | **14%** |

| | | | |
|---|---|---|---|
| 6.1 | I/O Devices, Device Controllers, Direct Memory Access | | |
| 6.2 | Principles of Input/output S/W: Goals of The I/O S/W, Interrupt Handler, Device Driver, Device Independent | | |
| 6.3 | I/O Software Disks: RAID levels, Disks Arm Scheduling Algorithm, Error Handling. | | |
| **7.** | **File Systems** | **06 Hours** | **15%** |
| 7.1 | File Naming, File Structure, File Types, File Access, File Attributes, File Operations, Memory Mapped Files | | |
| 7.2 | Directories: Hierarchical Directory System, Pathnames, Directory Operations, | | |
| 7.3 | File System Implementation, Contiguous Allocation, Linked List Allocation, Linked List Using Index, Inodes | | |
| **8.** | **Unix/Linux File System** | **04 Hours** | **10%** |
| 8.1 | Buffer Cache, Inodes, The system calls: ialloc, ifree, namei, alloc and free | | |
| 8.2 | Mounting and Unmounting, files systems, Network File systems | | |
| 8.3 | EXT file system in Linux | | |

**Course Outcome (COs):**

At the end of the course, the students will be able to

| | |
|---|---|
| CO1 | Visualize and understand Operating system functionality and working of OS. Understanding of functionality, services of operating system and differentiate between different types of OS. |
| CO2 | Define thread and process. Visualize how processes and threads are managed by the operating systems. Simulate and analyze various process scheduling algorithms. Explain and analyze different Inter process communication techniques |
| CO3 | Describe deadlock and classify detection, recovery, prevention and avoidance algorithms. Test scenarios to report deadlock |
| CO4 | Compare and evaluate various memory management schemes. Simulate and analyze Memory management algorithms. Identify and describe the role of I/O devices. |
| CO5 | Understand the file systems. Understand the secondary storage and simulate disk scheduling algorithm. |
| CO6 | Understand the basic commands of Linux file systems |

**Course Articulation Matrix:**

| | PO01 | PO02 | PO03 | PO04 | PO05 | PO06 | PO07 | PO08 | PO09 | PO10 | PO11 | PO12 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 3 | - | - | - | 2 | - | - | - | - | - | - | 1 | - | - |
| CO2 | 3 | 2 | - | - | 2 | - | - | - | - | - | - | - | 1 | - |
| CO3 | 3 | 2 | - | - | 2 | - | - | - | - | - | - | - | 1 | - |

| | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO4 | 3 | 2 | - | - | 2 | - | - | - | - | - | - | - | 1 | - |
| CO5 | 3 | 2 | - | - | 2 | - | - | - | - | - | - | - | 1 | - |
| CO6 | 3 | 2 | - | - | 2 | - | - | - | - | - | - | 1 | 1 | - |

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial

(High) If there is no correlation, put "-"

## Recommended Study Material:

- **Text Books:**

  1. Modern Operating Systems -By Andrew S. Tanenbaum, Third Edition PHI
  2. Operating System Concepts Avi Silberschatz, Peter Baer Galvin, Greg Gagne, Ninth Edition, Wiley

- **Reference Books:**

  1. Operating Systems, D.M. Dhamdhare, TMH
  2. Operating Systems Internals and Design Principles , William Stallings , Seventh Edition, Prentice Hall
  3. Unix System Concepts & Applications, Sumitabha Das, TMH
  4. Unix Shell Programming, Yashwant Kanitkar, BPB Publications

## AIML209: Probabilistic Modelling and Reasoning with Python

**Credits and Hours:**

| Teaching Scheme | Theory | Practical | Tutorial | Total | Credit |
|---|---|---|---|---|---|
| Hours/week | 3 | 2 | - | 5 | **4** |
| Marks | 100 | 50 | - | 150 | |

**Outline of the Course:**

| Sr. No. | Title of the unit | Minimum number of hours |
|---|---|---|
| 1. | Introduction to Statistics | 09 |
| 2. | Probability Theory | 12 |
| 3. | Point Estimations | 12 |
| 4. | Test of Statistical Hypothesis and p-values | 12 |
| | Total hours (Theory) : | 45 |
| | Total hours (Lab) : | 30 |
| | Total hours : | 75 |

**Detailed Syllabus:**

| | | | |
|---|---|---|---|
| **1.** | **Introduction to Statistics** | **09 Hours** | **19%** |
| | Introduction to Statistics. Role of statistics in scientific methods, current applications of statistics, Scientific data gathering: Sampling techniques, scientific studies, observational studies, data management. Data description: Displaying data on a single variable (graphical methods, measure of central tendency, measure of spread), displaying relationship between two or more variables, measure of association between two or more variables. | | |
| **2.** | **Probability Theory** | **12 Hours** | **27%** |
| | Probability Theory: Sample space and events, probability, axioms of probability, independent events, conditional probability, Bayes' theorem. Random Variables: Discrete and continuous random variables. Probability distribution of | | |

| | | | |
|---|---|---|---|
| | discrete random variables, binomial distribution, poisson distribution. Probability distribution of continuous random variables, The uniform distribution, normal (gaussian) distribution, exponential distribution, gamma distribution, beta distribution, t-distribution, $\chi$ " distribution. Expectations, variance and covariance. Probability Inequalities. Bivariate distributions | | |
| **3.** | **Point Estimations** | **12 Hours** | **27%** |
| | Point Estimations: Methods of finding estimators, method of moments, maximum likelihood estimators, bayes estimators. Methods of evaluating estimators, mean squared error, best unbiased estimator, sufficiency and unbiasedness Interval Estimations: Confidence interval of means and proportions, Distribution free confidence interval of percentiles | | |
| **4.** | **Test of Statistical Hypothesis and p-values** | **12 Hours** | **27%** |
| | Tests about one mean, tests of equality of two means, test about proportions, p-values, likelihood ratio test, Bayesian tests Bayesian Statistics: Bayesian inference of discrete random variable, Bayesian inference of binomial proportion, comparing Bayesian and frequentist inferences of proportion, comparing Bayesian and frequentist inferences of mean Univariate Statistics using Python: Mean, Mode. Median, Variance, Standard Deviation, Normal Distribution, t-distribution, interval estimation, Hypothesis Testing, Pearson correlation test, ANOVA F-test | | |

**Course Outcome (COs):**

At the end of the course, the students will be able to:

| | |
|---|---|
| CO1 | Calculate the singular value decomposition of a given matrix and principal components of a given covariance data matrix for reducing the dimensions |
| CO2 | Apply moments, mean, variance, skewness and kurtosis of Gaussian Distributions in the geometrical analysis of a data set which follows Gaussian distributions. |
| CO3 | Make use of decision theory and estimation statistics, EM algorithm in finding maximum likelihood parameters of a statistical model |
| CO4 | Interpret the role of the log likelihood function and maximum likely hood estimate in determining the eatimates of Binomial,Poisson, Normal distribution parameters. |

**Course Articulation Matrix:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 3 | - | 2 | - | - | - | - | - | - | - | - | - | 3 | - |
| CO2 | - | 2 | - | - | - | - | - | - | - | - | - | - | 3 | - |
| CO3 | - | 3 | 3 | 3 | - | - | - | - | - | - | - | - | 2 | - |
| CO4 | - | 1 | - | 1 | 2 | - | - | - | - | - | - | - | 2 | - |
| CO5 | 2 | 2 | 2 | 2 | - | - | - | - | - | - | - | 2 | 3 | - |
| CO6 | 2 | - | - | - | - | - | - | - | - | - | - | - | 2 | - |

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

**Recommended Study Material:**

- ❖ **Text book:**
  1. Achim Klenke, (2014), Probability Theory A Comprehensive Course Second Edition, Springer, ISBN 978-1-4471-5360.

❖ **Reference book:**

1. Christian Heumann, Michael Schomaker Shalabh (2016), Introduction to Statistics and Data Analysis With Exercises, Solutions and Applications in R, Springer International Publishing, ISBN 978-3-319-46160-1
2. Douglas C. Montgomery, (2012), Applied Statistics and Probability for Engineers, 5th Edition, , Wiley India, ISBN: 978-8-126-53719-8.
3. S. C. Gupta, V. K. Kapoor, “Fundamentals of Mathematical Statistics”, S. Chand & Co., 12th Edition, 2016
4. Richard Arnold Johnson, Irwin Miller, John E. Freund, “Probability and Statistics for Engineers”, Prentice Hall, 8th Edition, 2013.
5. B. S. Grewal, “Higher Engineering Mathematics”, Khanna Publishers, 43rd Edition, 2012.

❖ **Web material:**

1. https://www.heavy.ai/learn/data-science
2. https://www.geeksforgeeks.org/data-science-process/
3. https://monkeylearn.com/machine-learning/
4. https://www.harness.io/blog/ai-ml-introduction

## AIML210: R PROGRAMMING FOR DATA SCIENCE AND DATA ANALYSIS

**Credits and Hours:**

| Teaching Scheme | Theory | Practical | Tutorial | Total | Credit |
|---|---|---|---|---|---|
| Hours/week | 3 | 2 | - | 5 | **4** |
| Marks | 100 | 50 | - | 150 | |

**Outline of the Course:**

| Sr. No. | Title of the unit | Minimum number of hours |
|---|---|---|
| 1. | Getting Started with R and R Workspace | 09 |
| 2. | Basic Objects and Basic Expressions | 12 |
| 3. | Working with Basic Objects and Strings | 12 |
| 4. | Working with Data – Visualize and Analyze Data | 12 |
| | Total hours (Theory) : | 45 |
| | Total hours (Lab) : | 30 |
| | Total hours : | 75 |

**Detailed Syllabus:**

| | | | |
|---|---|---|---|
| **1.** | **Getting Started with R and R Workspace** | **09 Hours** | **19%** |
| | Introducing R, R as a programming Language, the need of R, Installing R, RStudio, RStudio's user interface, console, editor, environment pane, history pane, file pane, plots pane, package pane, help and viewer pane R Workspace, R's working directory, R Project in R Studio, absolute and relative path, Inspecting an Environment, Inspect existing Symbols, View the structure of object, Removing symbols, Modifying Global Options, Modifying warning level, Library of Packages, Getting to know a package, Installing a Package from CRAN, Updating Package from CRAN, Installing package from online repository, Package Function, Masking and name conflicts | | |
| **2.** | **Basic Objects and Basic Expressions** | **12 Hours** | **27%** |

| | | | |
|---|---|---|---|
| | Vectors, Numeric Vectors, Logical Vectors, Character Vectors, subset vectors, Named Vectors, extracting element, converting vector, Arithmetic operators, create Matrix, Naming row and columns, subsetting matrix, matrix operators, creating and subsetting an Array, Creating a List, extracting element from list, subsetting a list, setting value, creating a value of data frame, subsetting a data frame, setting values, factors, useful functions of a data frame, loading and writing data on disk, creating a function, calling a function, dynamic typing, generalizing a function. Assignment Operators, Conditional Expression, using if as expression and statement, using if with vectors, vectorized if: ifelse, using switch, using for loop, nested for loop, while loop | | |
| **3.** | **Working with Basic Objects and Strings** | **12 Hours** | **27%** |
| | Working with object function, getting data dimensions, reshaping data structures, iterating over one dimension, logical operators, logical functions, dealing with missing values, logical coercion, math function, number rounding functions, trigonometric functions, hyperbolic functions, extreme functions, finding roots, derivatives and integration, Statistical function, sampling from a vector, Working with random distributions, computing summary statistics, covariance and correlation matrix, printing string, concatenating string, transforming text, Formatting text, formatting date and time, formatting date and time to string, finding string pattern, using group to extract data, reading data | | |
| **4.** | **Working with Data – Visualize and Analyze Data** | **12 Hours** | **27%** |
| | Reading and Writing Data, importing data using built-in function, READR package, export a data frame to file, reading and writing Excel worksheets, reading and writing native data files, loading built-in data sets, create scatter plot, bar chart, pie chart, histogram and density plots, box plot, fitting linear model and regression tree. | | |

**Course Outcome (COs):**

At the end of the course, the students will be able to:

| | |
|---|---|
| CO1 | Understand critical R programming concepts. |
| CO2 | Demonstrate how to install and configure R studio. |
| CO3 | Explain the use of data structure and loop functions. |
| CO4 | Analyse data and generate reports based on the dataset. |

**Course Articulation Matrix:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 3 | - | 2 | - | - | - | - | - | - | - | - | - | 3 | - |
| CO2 | - | 2 | - | - | - | - | - | - | - | - | - | - | 3 | - |
| CO3 | - | 3 | 3 | 3 | - | - | - | - | - | - | - | - | 2 | - |
| CO4 | - | 1 | - | 1 | 2 | - | - | - | - | - | - | - | 2 | - |
| CO5 | 2 | 2 | 2 | 2 | - | - | - | - | - | - | - | 2 | 3 | - |
| CO6 | 2 | - | - | - | - | - | - | - | - | - | - | - | 2 | - |

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put "-"

**Recommended Study Material:**

- **Text book:**
  1. Hands-On Programming with R by Garrett Grolemund
- **Reference book:**
  1. R for Data Science by Hadley Wickham & Garrett Grolemund
- **Web material:**
  1. https://www.amazon.in/dp/1491910399?tag=guru99-21&geniuslink=true
  2. https://leanpub.com/rprogramming
  3. http://www.gvpcew.ac.in/Material/IT/2%20IT%20-%20UNIT-1%20Start%20Learning%20R.pdf

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# AIML211: PROJECT-I

## Credits and Hours:

| Teaching Scheme | Theory | Practical | Tutorial | Total | Credit |
|---|---|---|---|---|---|
| Hours/week | 0 | 4 | - | 4 | **2** |
| Marks | 0 | 100 | - | 100 | |

**Pre-requisite courses:**

- Programming Language, Software Engineering.

## Outline of the Course:

- Student at the beginning of a semester may be advised by his/her supervisor (s) for recommended courses.
- Students will work together in a team (at most three) with any programming language.
- Students are required to get approval of project definition from the department.
- After approval of project definition students are required to report their project work on weekly basis to the respective internal guide.
- Project will be evaluated at least once per week in laboratory Hours during the semester and final submission will be taken at the end of the semester as a part of continuous evaluation.
- Project work should include whole SDLC of development of software / hardware system as a solution of particular problem by applying principles of Software Engineering.
- Students have to submit project with following listed documents at the time of final submission.
  a. Final Project Report
  b. Project Setup file with Source code
  c. Project Presentation (PPT)
- A student has to produce some useful outcome by conducting experiments or projectwork.

Total hours (Theory) : 00
Total hours (Lab) : 30
Total hours : 30

**Course Outcome (COs):**

At the end of the course, the students will be able to

| | |
|---|---|
| CO1 | Create enhanced employment by moulding the students with higher technical skill. |
| CO2 | Promote creative thinking, to provide hands on actual technology. |
| CO3 | Handle software project and get use to software development processes. |
| CO4 | Correlate knowledge of different subjects and apply theoretical knowledge to implement project for identified problem. |
| CO5 | Write technical report and deliver presentation by applying different visualization tools and evaluation metrics. |
| CO6 | Improve communication and presentation skill. |

**Course Articulation Matrix:**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | - | 3 | - | - | - | - | - | - | - | - | 1 | - | - | - |
| CO2 | - | - | - | - | - | - | - | - | - | - | - | - | - | - |
| CO3 | - | - | - | - | 2 | - | - | - | 2 | - | - | - | - | - |
| CO4 | 3 | - | - | - | - | - | - | - | - | - | - | - | 3 | - |
| CO5 | - | - | 1 | - | 3 | 2 | - | - | - | 3 | - | - | - | 2 |
| CO6 | - | - | - | - | - | - | 3 | 1 | - | 3 | - | 2 | - | 3 |

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put "-"

**Recommended Study Material:**

- ❖ **Reference book:**
  1. Books, Magazines ,Journals & online course platforms of related topics
- ❖ **Web material:**
  1. www.ieeexplore.ieee.org
  2. www.sciencedirect.com
  3. www.elsevier.com
  4. https://www.udemy.com/
  5. https://www.udacity.com/
  6. https://nptel.ac.in/course.html
  7. https://www.futurelearn.com/

- **Software:**
    1. ASP.NET
    2. PYTHON/MATLAB
    3. PHP
    4. ANDROID/IOS